Jay Barrett

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Education

Cornell University Ithaca, NY
Bachelor of Arts in Computer Science, Minor in Business *Expected May 2026, GPA: 3.2*

Relevant Experience

Consultant - Digital Workplace Services Team June 2025 – August 2025
Kyndryl *Rye Brook, NY*

- Managed deployment and configuration of Python-based data pipeline extensions for Kyndryl Bridge AIOps platform, processing ServiceNow data through Apache Kafka streams to Elasticsearch visualization dashboards
- Secured multi-environment access (dev/staging/production) and configured OAuth authentication for enterprise-grade deployment pipeline supporting IT ops with CSAT score optimization and MTTR reduction goals
- Led AI web-agent integration technical analysis of 500+ enterprise applications as part of 5,800-application rationalization initiative targeting 10-30% portfolio reduction and $10M-$50M+ cost optimization

Data Annotation & LLM Evaluation - Independent Contractor May 2024 – Present
Scale AI *Remote*

- Senior Reviewer making final approvals on 140+ enterprise LLM artifacts weekly, maintaining 90%+ QA standards
- Audited training datasets for client delivery, maintaining <2% defect rate on enterprise AI model deployments
- Contributed SFT/RLHF training data for production AI systems deployed to enterprise clients

Projects

Local LLM Fine-Tuning Pipeline | *PyTorch, LoRA, Transformers, Docker, Claude Code* August 2025

- Fine-tuned Qwen2.5-7B locally using LoRA on RTX 3080 Ti, achieving 86% XML structure consistency (14 percentage points above baseline) for Model Context Protocol applications
- Leveraged Claude Code's autonomous development capabilities to generate 450 high-quality training examples overnight, achieving 98.7% schema validity with minimal human intervention
- Implemented complete MLOps pipeline with 4-bit quantization, flash attention optimization, and containerized deployment using Docker and NVIDIA CUDA

Minecraft Ollama Integration | *Go, Docker, Ollama, RAG, XML Processing, Bedrock API* July 2025

- Built real-time AI system processing all Minecraft chat messages through Go MCP server with rate-limited queues, RAG-powered memory, and autonomous 3-minute commentary triggers
- Integrated qwen2.5:14b-instruct via Ollama with vector database for semantic search, XML command parsing, and persistent conversation history across server restarts
- Deployed containerized architecture on MacBook M4 with behavior pack integration, server console injection, and Docker Compose orchestration

Database Management System | *Java, SQL, JSqlParser, NIO, JUnit* November 2024

- Engineered Database Management System with cost-based optimizer, B-tree indexing, and dynamic join selection
- Implemented three join algorithms (TNLJ, BNLJ, SMJ) with selectivity-based optimization, achieving 60% query performance improvement

Research

HAVER Algorithm Research June 2024 – October 2024
University of Arizona *Tucson, AZ*

- Co-authored *HAVER: Instance-Dependent Error Bounds for Maximum Mean Estimation and Applications to Q-Learning and Monte Carlo Tree Search*, leading to successful ArXiv publication
- Implemented Monte Carlo Tree Search with Upper Confidence bounds applied to Trees (MCTS-UCT) algorithm from scratch in Python using OpenAI Gym environments
- Developed reward and loss heuristics based on established research papers, contributing to successful AISTATS 2025 submission

Technical Skills

Python, Java, JavaScript, TypeScript, C, SQL, Golang, React, Next.js, Angular, Node.js, Express, Flask, PostgreSQL, MongoDB, Redis, Kafka, Docker, Kubernetes, AWS, Azure, GCP, Git, CI/CD, REST, GraphQL, Machine Learning, AI, NLP, LLM, PyTorch, TensorFlow, LoRA, Transformers, ETL, Data Pipeline, MLOps, Serverless, Linux, Event-driven Architecture, Load Balancing, Logging, Microservices, OAuth, RAG, Airflow, TDD, Agile, Kanban, Scrum, Django